BMI Calculator

A native Android calculator that validates height and weight, computes BMI, and displays the corresponding adult category.

Assignment coverage

Inputs and validation - Height: 50-300 cm. Weight: 10-700 kg. Empty, zero, negative, malformed and nonfinite inputs are rejected. Arabic digits and decimal commas are supported.

Calculation and category - BMI = weight in kg / height in metres squared. Categories use the unrounded value: below 18.5, below 25, below 30, or 30 and above.

Result handling - The result uses two decimal places. Input edits clear stale results; Reset clears the form. The calculated result survives Activity recreation.

Run and review

Open the repository root in Android Studio. Install SDK 35 and Build Tools 35.0.0; use JDK 17 or 21. Sync Gradle, then run the bmi module on Android 8.0 / API 26 or later.

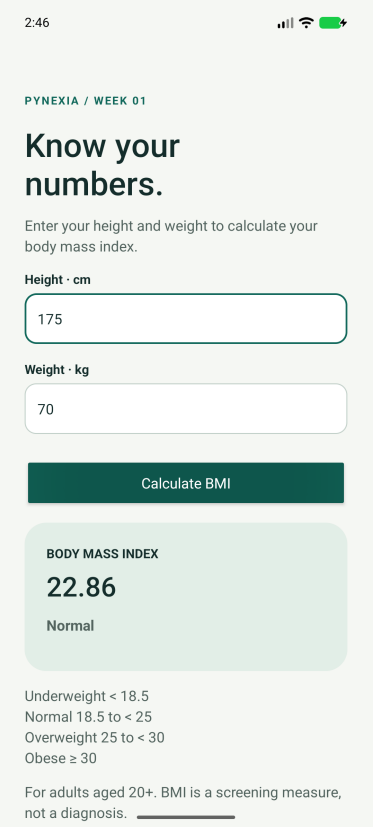
1. Enter height 175 cm and weight 70 kg, then tap Calculate BMI.
2. Confirm the result is 22.86 with the Normal category.
3. Change an input: the old result disappears until recalculation.
4. Try empty fields, zero or malformed input, then Reset. Rotate after a valid calculation to check restoration.

Scope

Categories apply to adults aged 20 and over. This is an educational screening calculator, not a diagnosis. Input ranges are guardrails rather than clinical cutoffs.

Actual app screens

Captured from the tested native Android application, not browser mockups.



175 cm + 70 kg = 22.86, Normal

Executed checks

JVM tests passed: 4 | Device tests passed: 2 | Lint errors: 0 | Lint warnings: 6

Four JVM tests cover calculation, category boundaries, localized parsing and invalid values. Two device tests cover calculation, clearing/resetting and Activity recreation.

Debug APK built and signature verified. Tests ran on a Pixel 8 emulator, Android 17 / API 37 (37.1 image), x86_64. Zero test failures, errors or skips. No physical-device or API 26 execution. Lint warnings are retained without suppression; inspect verification/ for the findings.

Reproduce

.\gradlew.bat :bmi:assembleDebug :bmi:testDebugUnitTest :bmi:lintDebug

.\gradlew.bat :bmi:connectedDebugAndroidTest

The second command requires a running emulator or connected device. APKs are debug-signed evaluation builds.

Reference: https://www.cdc.gov/bmi/adult-calculator/bmi-categories.html

Explanation & submission

يحسب التمرين مؤشر كتلة الجسم من الطول بالسنتيمتر والوزن بالكيلوغرام، ثم يعرض النتيجة وتصنيفها. يفحص المدخلات ويدعم الأرقام العربية والفاصلة العشرية. يعتمد التصنيف على القيمة الأصلية قبل التقريب، وتمسح النتيجة السابقة عند تعديل المدخلات.

للتجربة: أدخل طولا قدره ١٧٥ سم ووزنا قدره ٧٠ كغم. النتيجة المتوقعة ٢٢٫٨٦ ضمن المجال الطبيعي. جرب بعد ذلك قيمة صفرية أو حقلا فارغا للتحقق من رسائل الخطأ. التصنيفات للبالغين بعمر ٢٠ سنة فأكثر.

Repository URL

https://github.com/ENGEyad/pynexia-week1-bmi-calculator

Live Demo / Hosted URL

https://engeyad.github.io/pynexia-week1-bmi-calculator/

Hosted project showcase with real screenshots and APK/PDF downloads. It is not an in-browser Android emulator.

Attach File / Archive

BMI-Calculator.zip - includes source, APK, screenshots, verification and this report; below the 10 MB limit. PDF may also be attached separately. APK is inside the ZIP because the supplied form does not list APK as a direct attachment type.

Notes / Implementation Remarks (1326 / 2000 characters)

Native Android BMI Calculator built with Java and XML. It validates height in centimetres and weight in kilograms, calculates BMI as weight/(height in metres squared), and displays the value and Underweight, Normal, Overweight or Obese category. Classification uses the unrounded result. Decimal commas and Arabic digits are supported. Editing inputs clears the previous result; Reset clears the form. The calculated result survives Activity recreation.

Run: open the source folder in Android Studio, sync Gradle, and run the bmi module on Android 8.0/API 26 or later. Enter 175 cm and 70 kg, then tap Calculate BMI: expected result 22.86, Normal. Empty, zero, malformed and out-of-range inputs show validation errors. Accepted ranges are 50-300 cm and 10-700 kg. Adult BMI categories apply to ages 20+; this is an educational screening calculator, not a diagnosis.

Validation: debug APK built successfully; 4 JVM tests and 2 instrumentation tests passed. Device testing used a Pixel 8 emulator (Android 17/API 37). Lint completed with zero errors; warnings are documented. The attached ZIP includes source, Gradle Wrapper, APK, screenshots and verification reports.

Hosted URL: project showcase with actual screenshots and APK/PDF downloads; not a browser emulator. The ZIP also includes a bilingual implementation report.